

Quadratic Functions — Graphs & Properties

Full Solutions & Explanations

Difficulty Mix: Easy (Q1–14) | Medium (Q15–30) | Hard (Q31–40) · Total: 40 Questions

Q1. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Vertex x-coordinate: $x = -b/(2a) = -(-6)/(2 \cdot 1) = 6/2 = 3$.

Vertex y-coordinate: $y = (3)^2 - 6(3) + 8 = 9 - 18 + 8 = -1$.

Vertex = (3, -1). ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $x = 3$, $y = 9 - 18 + 8 = -1$.

B) $y = 9 - 18 + 8 = -1$, not +1. Sign error on the constant.

C) $x = -b/(2a) = +3$, not -3. Sign error in formula.

D) (6,8) are the coefficients b and c, not the vertex coordinates.

■ **TRAP ALERT:** Students often pick up b and c as the vertex. Always apply $x = -b/(2a)$, then substitute back.

■ **KEY INSIGHT:** Vertex $x = -b/(2a)$. Then substitute into the equation for the y-value.

■ **FAST METHOD:** Complete the square: $x^2 - 6x + 8 = (x - 3)^2 - 1$. Vertex = (3, -1) by inspection.

Q2. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

The y-intercept is the value of y when $x = 0$.

$y = 2(0)^2 - 5(0) + 3 = 3$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is the leading coefficient a, not the y-intercept.

B) This is the coefficient b, not the y-intercept.

C) ✓ CORRECT — CORRECT. Setting $x = 0$ gives $y = 3$ (the constant term c).

D) The y-intercept is not zero; the parabola does not pass through the origin.

■ **TRAP ALERT:** The y-intercept of $y = ax^2 + bx + c$ is ALWAYS c — just read the constant term.

■ **KEY INSIGHT:** For any polynomial, the y-intercept = $f(0)$ = the constant term.

■ **FAST METHOD:** y-intercept = $c = 3$. No calculation needed.

Q3. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

The coefficient of x^2 is -1 (negative).

When $a < 0$, the parabola opens DOWNWARD.

Answer: Downward because the coefficient of x^2 is negative. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Upward requires $a > 0$. Here $a = -1 < 0$.

B) ✓ CORRECT — CORRECT. Negative leading coefficient → opens downward.

C) Direction is determined by sign of a , not c .

D) Reason is wrong even though direction is right.

■ **TRAP ALERT:** Direction of opening depends ONLY on the sign of a (coefficient of x^2).

■ **KEY INSIGHT:** $a > 0$ → opens up (minimum). $a < 0$ → opens down (maximum).

■ **FAST METHOD:** Check sign of x^2 term: negative → downward. Done.

Q4. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

Factor: $y = x^2 - 5x + 6 = (x-2)(x-3)$.

x -intercepts: $x = 2$ or $x = 3$.

From the options, $x = 2$ is listed. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $1^2 - 5(1) + 6 = 2 \neq 0$.

B) ✓ CORRECT — CORRECT. $(2-2)(2-3) = 0$ ✓.

C) $25 - 25 + 6 = 6 \neq 0$.

D) $36 - 30 + 6 = 12 \neq 0$.

■ **TRAP ALERT:** The factors of 6 that add to -5 are -2 and -3. Students often try +2, +3.

■ **KEY INSIGHT:** Quick factor check: plug each answer into the equation. If result = 0, it's a root.

■ **FAST METHOD:** Look for factor pairs of 6 that sum to -5: $(-2)(-3)$. So roots are $x=2$ and $x=3$.

Q5. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

Axis of symmetry: $x = -b/(2a) = -(-12)/(2*3) = 12/6 = 2$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Sign error: $-(-12)/(6) = +2$, not -2.

B) ✓ CORRECT — CORRECT. $x = 12/6 = 2$.

C) Divided b by a instead of 2a: $12/3 = 4$.

D) This is just the value of b, not $-b/(2a)$.

■ **TRAP ALERT:** The formula is $x = -b/(2a)$. Don't forget to divide by 2a, not just a.

■ **KEY INSIGHT:** Axis of symmetry is always $x = -b/(2a)$. The axis passes through the vertex.

■ **FAST METHOD:** $x = -(-12)/(2 \times 3) = 12/6 = 2$.

Q6. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

$$f(x) = x^2 - 4 = (x-2)(x+2).$$

x-intercepts: $x = 2$ and $x = -2$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $x^2=4$ gives $x=\pm 2$, not just $x=4$.

B) These would come from $x^2=16$, not $x^2=4$.

C) ✓ CORRECT — CORRECT. $x^2=4 \rightarrow x=\pm 2$.

D) There are two roots; don't ignore the negative one.

■ **TRAP ALERT:** $x^2=4$ has TWO solutions: $x=2$ AND $x=-2$. Never forget the negative square root.

■ **KEY INSIGHT:** Difference of squares: $x^2-4 = (x-2)(x+2) \rightarrow$ roots at ± 2 .

■ **FAST METHOD:** Set $x^2=4$, take square root: $x=\pm 2$.

Q7. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Vertex form: $y = (x-h)^2 + k$ has vertex (h, k) .

Here $h=5$, $k=3$. Vertex = $(5, 3)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. In $y=(x-5)^2+3$, $h=5$ (not -5), $k=3$.

B) The shift is $x-5$, so $h=+5$, not -5.

C) $k=+3$, not -3.

D) Both signs are wrong.

■ **TRAP ALERT:** In $(x-h)^2$, the vertex x-coordinate is $+h$ (not -h). The minus sign is part of the form.

■ **KEY INSIGHT:** $y=(x-h)^2+k$: vertex is (h,k) . The subtraction inside means positive h.

■ **FAST METHOD:** $y=(x-5)^2+3 \rightarrow$ vertex= $(5,3)$. Read h and k directly.

Q8. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

The parabola opens upward (minimum exists).

For a parabola in vertex form $y=a(x-h)^2+k$ with $a>0$, the minimum value is k .

Here the vertex is $(2,-7)$, so the minimum value is -7 . ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is the x-coordinate of the vertex, not the minimum y-value.

B) ✓ CORRECT — CORRECT. The minimum value of f is the y-coordinate of the vertex.

C) The minimum is -7 (negative), not $+7$.

D) An upward-opening parabola always has a minimum at its vertex.

■ **TRAP ALERT:** The minimum VALUE of the function is the y-coordinate of the vertex, not the x-coordinate.

■ **KEY INSIGHT:** Minimum/maximum VALUE = y-coordinate of vertex. Location = x-coordinate of vertex.

■ **FAST METHOD:** Vertex $(2,-7)$, opens upward → minimum value = -7 (the y-coordinate).

Q9. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

A vertex at the origin means the vertex is $(0,0)$.

$y = x^2$ has vertex $(0,0)$ since it's $y = (x-0)^2 + 0$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Vertex at $(0,3)$, not origin.

B) Vertex at $(3,0)$, not origin.

C) ✓ CORRECT — CORRECT. Vertex at $(0,0)$.

D) Vertex at $(1/2, -1/4)$, not origin.

■ **TRAP ALERT:** 'Vertex at origin' means both $h=0$ AND $k=0$.

■ **KEY INSIGHT:** $y=x^2$ is the parent quadratic with vertex at the origin.

■ **FAST METHOD:** $y=x^2 = (x-0)^2+0$ → vertex $(0,0)$ ✓.

Q10. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Roots $x=-1$ and $x=3$ mean factors $(x+1)$ and $(x-3)$.

$y = (x+1)(x-3) = x^2-3x+x-3 = x^2-2x-3$. ✓

This matches the original equation.

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Roots -1 and 3 → factors $(x-(-1))(x-3) = (x+1)(x-3)$.

B) Gives roots $x=1$ and $x=-3$, not -1 and 3 .

C) Gives roots $x=-1$ and $x=-3$.

D) Gives roots $x=1$ and $x=3$.

■ **TRAP ALERT:** Root $x = r$ corresponds to factor $(x - r)$, not $(x + r)$. So root $-1 \rightarrow$ factor $(x+1)$.

■ **KEY INSIGHT:** If the roots are r and s , the factored form is $y = (x-r)(x-s)$.

■ **FAST METHOD:** Roots -1 and $3 \rightarrow (x-(-1))(x-3) = (x+1)(x-3)$.

Q11. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

$a > 0$ means the parabola opens upward.

An upward-opening parabola has a MINIMUM value (at its vertex).

Answer: C ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Downward requires $a < 0$.

B) Whether there are x -intercepts depends on the discriminant, not just a .

C) ✓ CORRECT — CORRECT. $a > 0 \rightarrow$ opens upward $\rightarrow$ minimum at vertex.

D) The vertex could be below, on, or above the x -axis.

■ **TRAP ALERT:** $a > 0$ guarantees a minimum value exists, but says nothing about x -intercepts or vertex position relative to x -axis.

■ **KEY INSIGHT:** Sign of a tells direction (up/down). Discriminant tells number of roots. Both matter independently.

■ **FAST METHOD:** $a > 0 \rightarrow$ opens up $\rightarrow$ has a minimum. That's C.

Q12. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

$y = x^2 + 6x + 9 = (x+3)^2 \rightarrow$ vertex form is $(x-(-3))^2+0$.

Vertex = $(-3, 0)$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $(x+3)^2=0$ at $x=-3$, so vertex is $(-3,0)$.

B) $x+3=0$ gives $x=-3$, not $+3$.

C) The $+9$ in x^2+6x+9 is absorbed into the perfect square; $k=0$.

D) $y(0)=9$ is the y -intercept, not the vertex.

■ **TRAP ALERT:** $x^2+6x+9 = (x+3)^2$ is a perfect square trinomial. The vertex is $(-3,0)$, sitting ON the x -axis.

■ **KEY INSIGHT:** A perfect square trinomial like $(x+h)^2$ touches the x-axis at exactly one point — its vertex.

■ **FAST METHOD:** $x^2+6x+9 = (x+3)^2 \rightarrow \text{vertex } (-3,0)$.

Q13. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Set $y=0$: $x^2+4=0 \rightarrow x^2=-4$.

$x^2=-4$ has NO real solutions (square of real number can't be negative).

The parabola has 0 x-intercepts. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $x^2=-4$ has no real solutions.

B) One intercept would require a perfect square; discriminant would be 0.

C) Two intercepts would require discriminant > 0 .

D) A parabola has at most 2 x-intercepts.

■ **TRAP ALERT:** Students may write $x^2=4$ and get $x=\pm 2$, forgetting the sign: $x^2+4=0 \rightarrow x^2=-4$.

■ **KEY INSIGHT:** Discriminant $b^2-4ac = 0-4(1)(4) = -16 < 0 \rightarrow \text{no real roots}$.

■ **FAST METHOD:** The vertex $(0,4)$ is above x-axis and parabola opens up $\rightarrow$ never crosses x-axis.

Q14. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

$f(x) = -(x-1)^2+9$. Since the coefficient of $(x-1)^2$ is -1 (negative), the parabola opens downward.

The maximum occurs at the vertex, where $x=1$.

$f(1) = -(1-1)^2+9 = 9$, the maximum value. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $x=-1$ is not the vertex of $-(x-1)^2+9$.

B) The vertex x-coordinate is 1, not 0.

C) ✓ CORRECT — CORRECT. Maximum occurs at vertex $x=1$ (where $(x-1)^2=0$, maximising the whole expression).

D) 9 is the maximum VALUE of f , not the x-value where it occurs.

■ **TRAP ALERT:** The question asks for the x-value at which the maximum occurs ($x=1$), not the maximum value itself (9).

■ **KEY INSIGHT:** In $-(x-h)^2+k$, the maximum value is k and it occurs at $x=h$.

■ **FAST METHOD:** Vertex at $x=1$ (from vertex form). That's where maximum occurs.

Q15. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

For exactly one real root, the discriminant = 0.

$$f(x) = x^2 - 8x + k \rightarrow \text{discriminant} = b^2 - 4ac = 64 - 4k.$$

$$\text{Set} = 0: 64 - 4k = 0 \rightarrow 4k = 64 \rightarrow k = 16. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) Discriminant = $64 - 16 = 48 > 0 \rightarrow$ two roots.

B) Discriminant = $64 - 32 = 32 > 0 \rightarrow$ two roots.

C) $\checkmark$ CORRECT — CORRECT. Discriminant = $64 - 64 = 0 \rightarrow$ exactly one root.

D) Discriminant = $64 - 256 < 0 \rightarrow$ no real roots.

■ **TRAP ALERT:** One real root requires discriminant = 0, not > 0 or < 0 .

■ **KEY INSIGHT:** Discriminant = $b^2 - 4ac$: > 0 two roots, $= 0$ one root, < 0 no real roots.

■ **FAST METHOD:** $b^2 = 4ac \rightarrow 64 = 4k \rightarrow k = 16$.

Q16. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$\text{Set } y = 0: 2(x-3)^2 - 8 = 0 \rightarrow (x-3)^2 = 4 \rightarrow x-3 = \pm 2.$$

$$x = 3 + 2 = 5 \text{ or } x = 3 - 2 = 1. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $\checkmark$ CORRECT — CORRECT. $3 \pm 2 = 1$ and 5 .

B) These would come from 3 ± 4 (wrong: $\pm \sqrt{4} = \pm 2$).

C) Asymmetric — both should be 3 ± 2 .

D) Asymmetric — both should be 3 ± 2 .

■ **TRAP ALERT:** $(x-3)^2 = 4 \rightarrow x-3 = \pm 2 \rightarrow x = 3 \pm 2$. Students sometimes write $x = \pm 2$ forgetting the -3 shift.

■ **KEY INSIGHT:** From vertex form, set = 0 and take square root: $x = h \pm \sqrt{-k/a}$.

■ **FAST METHOD:** $(x-3)^2 = 4 \rightarrow x-3 = \pm 2 \rightarrow x = 1$ or $x = 5$.

Q17. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

Second quadrant: $x < 0$ and $y > 0$.

Vertex of $y = (x+2)^2 + 3$ is $(-2, 3)$. $x = -2 < 0$ and $y = 3 > 0 \rightarrow$ second quadrant. $\checkmark$

WHY EACH OPTION IS / IS NOT CORRECT:

A) Vertex $(2, 3) \rightarrow$ first quadrant ($x > 0, y > 0$).

B) $\checkmark$ CORRECT — CORRECT. Vertex $(-2, 3) \rightarrow$ second quadrant.

C) Vertex (2,-3) → fourth quadrant.

D) Vertex (-2,-3) → third quadrant.

■ **TRAP ALERT:** Know all four quadrants: Q1(+,+), Q2(-,+), Q3(-,-), Q4(+,-). Match vertex signs.

■ **KEY INSIGHT:** From vertex form $y=(x-h)^2+k$, vertex=(h,k). Check quadrant from signs of h and k.

■ **FAST METHOD:** Second quadrant: $h<0$, $k>0$. Only option B has $h=-2$, $k=+3$.

Q18. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

$h(t) = -16t^2 + 64t + 5$. Max height at vertex: $t = -64/(2 \times -16) = -64/-32 = 2$.

$h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$ feet. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is $h(0)$, the initial height, not the maximum.

B) This is the coefficient of t, not the max height.

C) ✓ CORRECT — CORRECT. $h(2)=69$ ft.

D) $64 \times 2 = 128$ ignores the $-16t^2$ term.

■ **TRAP ALERT:** Evaluate h at the vertex $t=2$, not at $t=0$ or $t=64$.

■ **KEY INSIGHT:** For $h(t) = -16t^2 + vt + h$, max height = $h + v^2/64$. Or just find $t = -b/2a$, then evaluate.

■ **FAST METHOD:** $t=2$. $h(2) = -64 + 128 + 5 = 69$.

Q19. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

Tangent to x-axis means discriminant = 0.

$y = x^2 - 4x + c \rightarrow b^2 - 4ac = 16 - 4c = 0 \rightarrow c = 4$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $16 - 8 = 8 > 0 \rightarrow$ two x-intercepts, not tangent.

B) ✓ CORRECT — CORRECT. $16 - 16 = 0 \rightarrow$ tangent to x-axis.

C) $16 - 32 = -16 < 0 \rightarrow$ no x-intercepts.

D) $16 + 16 = 32 > 0 \rightarrow$ two x-intercepts.

■ **TRAP ALERT:** 'Tangent to x-axis' = discriminant equals 0 (one repeated root), not discriminant > 0 .

■ **KEY INSIGHT:** Tangent to x-axis $\leftrightarrow$ discriminant = 0 $\leftrightarrow$ the parabola's vertex sits exactly on the x-axis.

■ **FAST METHOD:** $b^2 = 4ac \rightarrow 16 = 4c \rightarrow c = 4$.

Q20. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

Vertex (2,-5) means $y=a(x-2)^2-5$.

Plug in (4,3): $3 = a(4-2)^2-5 = 4a-5 \rightarrow 4a=8 \rightarrow a=2$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $1 \times 4 - 5 = -1 \neq 3$.

B) ✓ CORRECT — CORRECT. $2 \times 4 - 5 = 3$ ✓.

C) $3 \times 4 - 5 = 7 \neq 3$.

D) $4 \times 4 - 5 = 11 \neq 3$.

■ **TRAP ALERT:** The vertex gives you h and k. You still need a second point to find a.

■ **KEY INSIGHT:** With vertex and one point: substitute the point into $y=a(x-h)^2+k$ and solve for a.

■ **FAST METHOD:** $3=a(2)^2-5 \rightarrow 4a=8 \rightarrow a=2$.

Q21. Correct Answer: D [MEDIUM]

STEP-BY-STEP SOLUTION:

Exactly one x-intercept $\rightarrow$ discriminant = 0.

$y = x^2+kx+9 \rightarrow$ discriminant $= k^2-36 = 0 \rightarrow k^2=36 \rightarrow k=6$ or $k=-6$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $9-36=-27<0 \rightarrow$ no real intercepts.

B) $36-36=0$ ✓ $\rightarrow$ one intercept. But so does $k=-6$.

C) $36-36=0$ ✓ $\rightarrow$ one intercept. But so does $k=6$.

D) ✓ CORRECT — CORRECT. $k^2=36$ has two solutions: $k=6$ and $k=-6$.

■ **TRAP ALERT:** The discriminant equation $k^2=36$ has TWO solutions. Don't stop at $k=6$.

■ **KEY INSIGHT:** When k appears squared in the discriminant condition, always consider $\pm$.

■ **FAST METHOD:** $k^2=4(1)(9)=36 \rightarrow k=\pm 6$. Both B and C apply $\rightarrow$ answer D.

Q22. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

Vertex x: $t = -12/(2 \times -2) = -12/-4 = 3$. ✓ (matches given $x=3$)

$f(3) = -2(9)+12(3)-10 = -18+36-10 = 8$. So $m=8$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $-18+36-10=8$.

B) This is $-c = -(-10)$, a common sign error.

C) This is the coefficient of x, not the vertex y-value.

D) This is $12 \times 3/2$ — partial calculation error.

■ **TRAP ALERT:** Don't forget the -10 constant: $-18+36=18$, then $18-10=8$.

■ **KEY INSIGHT:** Always substitute x-coordinate back into the original equation for the vertex y-value.

■ **FAST METHOD:** $-2(9)+36-10=-18+36-10=8$.

Q23. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

Two real solutions $\rightarrow$ discriminant > 0 : $b^2-4ac = 36-28 = 8 > 0$. ✓

Discriminant > 0 means the parabola CROSSES the x-axis $\rightarrow$ vertex is BELOW x-axis.

Since $a=1>0$ (opens up), the vertex must be below the x-axis. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Disc >0 + upward opening $\rightarrow$ vertex below x-axis.

B) On x-axis would require disc $=0$.

C) Above x-axis with disc >0 would require downward opening.

D) Leading coefficient is $+1$, so it opens upward.

■ **TRAP ALERT:** Discriminant > 0 means two x-intercepts, which for an upward parabola means vertex is BELOW x-axis.

■ **KEY INSIGHT:** Upward parabola: vertex below x-axis $\rightarrow$ 2 roots. On x-axis $\rightarrow$ 1 root. Above x-axis $\rightarrow$ 0 roots.

■ **FAST METHOD:** Disc $=36-28=8>0 \rightarrow$ crosses x-axis twice $\rightarrow$ vertex must be below x-axis.

Q24. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

The axis of symmetry is midway between symmetric points.

(0,4) and (2,4) have the same y-value, so they are symmetric about $x=1$.

Axis of symmetry: $x=1$. ✓

Verify: vertex at $x=1$. $y(1)=a-b(a-b?)$... quick check: (0,4),(2,4) equidistant from $x=1$.

WHY EACH OPTION IS / IS NOT CORRECT:

A) $x=0$ is one of the points, not the axis.

B) ✓ CORRECT — CORRECT. (0,4) and (2,4) are equidistant from $x=1$.

C) $x=2$ is one of the points, not the axis.

D) No symmetry evidence supports $x=3$.

■ **TRAP ALERT:** Two points with the SAME y-value on a parabola are symmetric. The axis is at their midpoint.

■ **KEY INSIGHT:** If $f(a)=f(b)$, then the axis of symmetry is $x=(a+b)/2$.

■ **FAST METHOD:** Same y-value at $x=0$ and $x=2 \rightarrow \text{axis} = (0+2)/2 = x=1$.

Q25. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

$$R(x) = -5x^2 + 200x. \text{ Vertex } x = -200/(2 \times -5) = -200/-10 = 20.$$

Maximum revenue at $x=\$20$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $R(10) = -500 + 2000 = 1500$. Not maximum.

B) ✓ **CORRECT — CORRECT.** Vertex at $x=20$. $R(20) = -5(400) + 200(20) = -2000 + 4000 = 2000$.

C) $R(40) = -5(1600) + 8000 = -8000 + 8000 = 0$.

D) $R(200)$ is far past the maximum.

■ **TRAP ALERT:** The vertex gives maximum of the quadratic. Don't confuse x (price) with $R(x)$ (revenue).

■ **KEY INSIGHT:** For revenue/profit maximisation, always find the vertex x -coordinate of the quadratic model.

■ **FAST METHOD:** $x = -b/(2a) = -200/(2 \times -5) = 20$.

Q26. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$f(x) = x^2 - 10x + 21 = (x-5)^2 - 4.$$

Minimum value is -4 (at vertex $x=5$, opens upward).

Range: $y \geq -4$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ **CORRECT — CORRECT.** Minimum = -4 at vertex $(5, -4)$.

B) Sign error: vertex y -value is -4 , not $+4$.

C) Would come from vertex $(-5, ?)$ calculation error.

D) Range is restricted for a parabola; not all real numbers.

■ **TRAP ALERT:** The range of an upward parabola is [minimum value, ∞). Find the minimum first.

■ **KEY INSIGHT:** Complete the square: $x^2 - 10x + 21 = (x-5)^2 - 4 \rightarrow \text{min} = -4 \rightarrow \text{range: } y \geq -4$.

■ **FAST METHOD:** Vertex: $x=5$, $y=25-50+21=-4$. Range: $y \geq -4$.

Q27. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

$$\text{Complete the square: } y = 3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

$$=3(x^2-2x+1-1)+5=3(x-1)^2-3+5=3(x-1)^2+2. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

- A) ✓ CORRECT — CORRECT. Vertex (1,2).
- B) Sign error: $-3+5=+2$, not -2 .
- C) Sign inside: x^2-2x makes $(x-1)^2$, not $(x+1)^2$.
- D) Vertex would be (2,5); not matching original.

■ **TRAP ALERT:** Factor out a from x^2 and x terms before completing the square. Then adjust the constant: $-a \times (\text{half coeff})^2$.

■ **KEY INSIGHT:** $y=3(x^2-2x)+5 \rightarrow$ add and subtract $3(1)$: $3(x-1)^2-3+5=3(x-1)^2+2$.

■ **FAST METHOD:** Vertex $x = -(-6)/(2 \times 3) = 1$. $y(1) = 3-6+5=2$. Vertex form: $3(x-1)^2+2$.

Q28. Correct Answer: A [MEDIUM]

STEP-BY-STEP SOLUTION:

Zeros at $x=-3$ and $x=7$, leading coefficient=2.

$$f(x)=2(x-(-3))(x-7)=2(x+3)(x-7). \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

- A) ✓ CORRECT — CORRECT. Roots $-3, 7 \rightarrow$ factors $(x+3), (x-7)$, scaled by 2.
- B) Gives roots $x=3$ and $x=-7$, not -3 and 7 .
- C) Missing the leading coefficient 2.
- D) Gives roots $x=-3$ and $x=-7$.

■ **TRAP ALERT:** Root $x=r \rightarrow$ factor $(x-r)$. Root $x=-3 \rightarrow$ factor $(x+3)$. Don't confuse sign.

■ **KEY INSIGHT:** $f(x)=a(x-r_1)(x-r_2)$ where a is the leading coefficient and r_1, r_2 are the zeros.

■ **FAST METHOD:** Zeros -3 and 7 , $a=2 \rightarrow 2(x+3)(x-7)$.

Q29. Correct Answer: D [MEDIUM]

STEP-BY-STEP SOLUTION:

Tangent to x -axis: discriminant=0.

$$y=x^2-bx+16 \rightarrow \text{disc}=b^2-4(1)(16)=b^2-64=0 \rightarrow b^2=64 \rightarrow b=8 \text{ or } b=-8. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

- A) $b=4$: $16-64=-48<0 \rightarrow$ no x -intercepts, not tangent.
- B) $b=-4$: same discriminant $= 16-64<0 \rightarrow$ not tangent.
- C) $b=8$ is correct, but $b=-8$ also works.
- D) ✓ CORRECT — CORRECT. $b^2=64 \rightarrow b=\pm 8$.

■ **TRAP ALERT:** The equation $b^2=64$ has TWO solutions. Always consider the negative root.

■ **KEY INSIGHT:** Tangent $\rightarrow \text{disc}=0 \rightarrow b^2=4ac$. Here $b^2=64 \rightarrow b=\pm 8$.

■ **FAST METHOD:** $b^2=4(1)(16)=64 \rightarrow b=\pm 8$.

Q30. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

Entirely above x-axis: vertex y-value > 0 (since $a=1>0$, opens up).

Vertex $y = k - \frac{b^2}{4a} = k - \frac{(-4)^2}{4} = k - 4 > 0 \rightarrow k > 4$. ✓

Actually: vertex of $y=x^2-4x+k$ is at $x=2$: $y=4-8+k=k-4$. For $k-4>0$: $k>4$.

WHY EACH OPTION IS / IS NOT CORRECT:

A) $k=3$: vertex $y=3-4=-1<0 \rightarrow$ not entirely above x-axis.

B) ✓ CORRECT — CORRECT. Vertex $y=k-4>0 \rightarrow k>4$.

C) This would put vertex below x-axis.

D) $k=4$ makes discriminant $=0 \rightarrow$ tangent (touches x-axis, not entirely above).

■ **TRAP ALERT:** $k>4$ strictly (not $k\geq 4$). At $k=4$ the parabola is tangent to x-axis (touches it).

■ **KEY INSIGHT:** Entirely above x-axis: vertex y-coordinate > 0 . At $k=4$, $\text{disc}=0$ (tangent, not entirely above).

■ **FAST METHOD:** Vertex $y = k-4$. For parabola entirely above x-axis: $k-4>0 \rightarrow k>4$.

Q31. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

Roots 1 and 7: vertex $x=(1+7)/2=4$.

$y=(x-1)(x-7)=x^2-8x+7$.

Minimum $= f(4)=16-32+7=-9$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $f(4)=16-32+7=-9$.

B) Off by 1; careful calculation gives -9.

C) This is $f(1)=f(7)=0$, the roots, not the minimum.

D) Sign error: minimum is -9, not +9.

■ **TRAP ALERT:** The minimum is the vertex y-value, not the average of the roots or the root values themselves.

■ **KEY INSIGHT:** Axis of symmetry = midpoint of roots $= (1+7)/2=4$. $\text{Min}=f(4)=-9$.

■ **FAST METHOD:** Vertex at $x=4$. $f(4)=16-32+7=-9$.

Q32. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

Original parabola: $y = x^2 + k$.

Reflected over the x-axis: $y = -(x^2 + k) = -x^2 - k$.

The reflected parabola passes through (2, -9):

$$-9 = -(2^2 + k) = -(4 + k)$$

$$\text{So } 4 + k = 9 \rightarrow k = 5. \checkmark$$

Verify: original $y = x^2 + 5$. Reflected: $y = -x^2 - 5$. At $x=2$: $y = -4 - 5 = -9 \checkmark$.

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is the y-coordinate of the reflected point, not k.

B) Sign error: $-(4+k)=-9$ gives $k=5$, not $k=-5$.

C) **✓ CORRECT — CORRECT.** $-(4+k)=-9 \rightarrow k=5$.

D) This comes from setting $-9=-(k)$ and ignoring the x^2 term.

■ **TRAP ALERT:** Reflecting $y=f(x)$ over the x-axis gives $y=-f(x)$. Here $y=-(x^2+k)$, not $y=x^2-k$.

■ **KEY INSIGHT:** Reflection over x-axis: negate the ENTIRE function. Then substitute the given point.

■ **FAST METHOD:** $-(4+k)=-9 \rightarrow 4+k=9 \rightarrow k=5$.

Q33. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$$f(x)=ax^2+bx+c.$$

$$f(0)=c=5.$$

$$f(1)=a+b+c=2. \rightarrow \text{This is } a+b+c \text{ directly!} = 2. \checkmark$$

$$\text{We can verify: } f(-1)=a-b+c=10. \ c=5: \ a-b=5. \ \text{And } a+b=2-5=-3.$$

$$\text{Adding: } 2a=2 \rightarrow a=1. \ b=-4. \ \text{Check } f(-1)=1+4+5=10 \checkmark.$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) **✓ CORRECT — CORRECT.** $a+b+c = f(1) = 2$ directly.

B) This is $c=f(0)$, not $a+b+c$.

C) This is $f(-1)=a-b+c$, not $a+b+c$.

D) Summing all three given values ($5+2+10$) — not what's asked.

■ **TRAP ALERT:** $a+b+c = f(1)$ directly. No need to find a, b, c individually.

■ **KEY INSIGHT:** For $f(x)=ax^2+bx+c$, $f(1)=a+b+c$ and $f(0)=c$. These shortcuts save enormous time.

■ **FAST METHOD:** $a+b+c = f(1) = 2$. One-step answer.

Q34. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

No real x-intercepts $\rightarrow$ discriminant < 0 .

$$y=x^2-2x+m \rightarrow \text{disc}=4-4m < 0 \rightarrow 4m > 4 \rightarrow m > 1. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $m < 1 \rightarrow \text{disc} > 0 \rightarrow$ two real intercepts.

B) $m = 1 \rightarrow \text{disc} = 0 \rightarrow$ one real intercept (tangent).

C) $\checkmark$ CORRECT — CORRECT. $\text{disc} = 4 - 4m < 0 \rightarrow m > 1$.

D) Too broad; includes $m = 0$ which gives $\text{disc} = 4 > 0$ (two roots).

■ **TRAP ALERT:** No real roots $\rightarrow \text{disc} < 0$ (strictly). At $m = 1$, $\text{disc} = 0$ means ONE root (tangent).

■ **KEY INSIGHT:** $\text{disc} < 0 \rightarrow$ no real roots. $\text{disc} = 0 \rightarrow$ one real root. $\text{disc} > 0 \rightarrow$ two real roots.

■ **FAST METHOD:** $4 - 4m < 0 \rightarrow m > 1$.

Q35. Correct Answer: D [HARD]

STEP-BY-STEP SOLUTION:

Tangent to x-axis: discriminant $= 0$.

$$y = 2x^2 - kx + 8 \rightarrow \text{disc} = k^2 - 4(2)(8) = k^2 - 64 = 0 \rightarrow k^2 = 64 \rightarrow k = 8 \text{ or } k = -8. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $16 - 64 = -48 < 0 \rightarrow$ no x-intercepts.

B) CORRECT for $k = 8$ alone, but $k = -8$ also works.

C) Also correct, but omits $k = 8$.

D) $\checkmark$ CORRECT — CORRECT. Both values make the discriminant zero.

■ **TRAP ALERT:** $k^2 = 64$ has TWO solutions. DSAT often includes the 'both' option — choose it when both values are valid.

■ **KEY INSIGHT:** Always solve $k^2 = \text{constant}$ to get $\pm\sqrt{\text{constant}}$. Present both solutions.

■ **FAST METHOD:** $k^2 = 4(2)(8) = 64 \rightarrow k = \pm 8$.

Q36. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

Vertex $(3, -4)$, so $f(x) = a(x-3)^2 - 4$.

Passes through $(5, 0)$: $0 = a(2)^2 - 4 = 4a - 4 \rightarrow a = 1$.

$$f(x) = (x-3)^2 - 4.$$

$$f(1) = (1-3)^2 - 4 = 4 - 4 = 0. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is the vertex y-value (minimum), not $f(1)$.

B) Calculation error: $(1-3)^2 = 4$, not 2.

C) ✓ CORRECT — CORRECT. $f(1)=(1-3)^2-4=4-4=0$.

D) Sign error: $4-4=0$, not 4.

■ **TRAP ALERT:** Find a from the second point first, then use the full equation to evaluate $f(1)$.

■ **KEY INSIGHT:** Two unknowns (a and k from vertex) need two pieces of info: the vertex gives (h,k) , and a second point gives a .

■ **FAST METHOD:** $a=1$ from $(5,0)$. $f(1)=(1-3)^2-4=0$.

Q37. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

Let the two numbers be x and $20-x$.

$$P=x(20-x)=-x^2+20x.$$

Maximum at $x=-20/(2\times-1)=10$.

$$P(10)=10\times10=100. \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) $P=19\times1=19$. Not the maximum.

B) $P=12\times8=96$. Close but not maximum.

C) ✓ CORRECT — CORRECT. $P(10)=10\times10=100$.

D) Would require both numbers to be 20, but sum would be $40\neq20$.

■ **TRAP ALERT:** The product is maximised when the two numbers are EQUAL (both = sum/2). This is a classic optimisation result.

■ **KEY INSIGHT:** AM-GM inequality: for fixed sum, product is maximised when both values are equal.

■ **FAST METHOD:** Product maximised at $x=20/2=10$. $P=100$.

Q38. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$$\text{Set equal: } x^2-3x-4 = x+k \rightarrow x^2-4x-(4+k) = 0.$$

For exactly one intersection: discriminant = 0.

$$\text{disc} = (-4)^2-4(1)(-(4+k)) = 16+4(4+k) = 32+4k.$$

$$\text{Set disc} = 0: 32+4k = 0 \rightarrow k = -8. \checkmark$$

$$\text{Verify: } x^2-4x+4 = 0 \rightarrow (x-2)^2 = 0 \rightarrow x = 2 \text{ (one repeated root).}$$

$$y = 2+(-8) = -6. \text{ Tangent point: } (2,-6). \checkmark$$

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. $\text{disc} = 32+4k = 0 \rightarrow k = -8$.

B) $\text{disc} = 32-20 = 12 > 0 \rightarrow$ two intersections.

C) $\text{disc} = 32+20 = 52 > 0 \rightarrow$ two intersections.

D) $\text{disc} = 32+32 = 64 > 0 \rightarrow$ two intersections.

■ **TRAP ALERT:** Set the two y-expressions equal, rearrange to $ax^2+bx+c=0$ form, then set discriminant=0.

■ **KEY INSIGHT:** For a line to be tangent to a parabola: discriminant of their intersection equation = 0.

■ **FAST METHOD:** $x^2-4x-(4+k)=0$. $\text{disc}=32+4k=0 \rightarrow k=-8$.

Q39. Correct Answer: A [HARD]

STEP-BY-STEP SOLUTION:

$g(x)=-x^2+6x-5=-(x-3)^2+4$. Vertex (3,4), opens downward.

On [0,6]: check endpoints and vertex.

$g(0)=-0+0-5=-5$.

$g(3)=4$ (maximum).

$g(6)=-36+36-5=-5$.

Range: $-5 \leq g \leq 4$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Min=-5 at $x=0$ and $x=6$, max=4 at $x=3$.

B) Max is 4, not 5.

C) Min is -5, not 0.

D) Both bounds are wrong.

■ **TRAP ALERT:** On a closed interval, always check BOTH endpoints AND the vertex.

■ **KEY INSIGHT:** For a restricted domain, the range is from the minimum to maximum, checking vertex and endpoints.

■ **FAST METHOD:** Vertex (3,4) is max. Endpoints: $g(0)=g(6)=-5$ is min. Range: [-5,4].

Q40. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

$f(x)=3x^2-12x+c=3(x-2)^2-12+c$.

Minimum value = $-12+c$.

Set $-12+c=7 \rightarrow c=19$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) This is the minimum value, not c.

B) $c=12$ gives $\text{min}=-12+12=0 \neq 7$.

C) ✓ CORRECT — CORRECT. $\text{min}=-12+c=7 \rightarrow c=19$.

D) $c=25$ gives $\text{min}=-12+25=13 \neq 7$.

■ **TRAP ALERT:** Complete the square to find the minimum in terms of c , then set equal to 7.

■ **KEY INSIGHT:** $3x^2 - 12x + c = 3(x-2)^2 + (c-12)$. Minimum $= c-12$. Set $c-12=7$ to find c .

■ **FAST METHOD:** Vertex $x=2$. $f(2)=12-24+c=c-12=7 \rightarrow c=19$.

TOPIC SUMMARY — QUADRATIC FUNCTIONS — GRAPHS & PROPERTIES

Digital SAT Math — Comprehensive Review

■ COMMON MISTAKES

- Using $x = -b/2a$ but forgetting to substitute back for the y-coordinate of the vertex.
- Confusing 'minimum value' (y-coordinate of vertex) with 'location of minimum' (x-coordinate).
- For discriminant problems: $\text{disc} > 0 \rightarrow$ two roots; $\text{disc} = 0 \rightarrow$ one root (tangent); $\text{disc} < 0 \rightarrow$ no real roots. Mixing these up is the #1 error.

■ MUST REMEMBER

- Vertex form: $y = a(x-h)^2 + k \rightarrow$ vertex (h,k) . In $(x-h)^2$, the vertex x is $+h$, not $-h$.
- Discriminant $b^2 - 4ac$: completely controls number of x-intercepts. Memorise all three cases.
- $f(1) = a + b + c$ and $f(-1) = a - b + c$ are instant shortcuts — no algebra needed.
- For product maximisation with fixed sum S : max product $= (S/2)^2$ (split equally).

■ EXPECTED ON DSAT

- Vertex identification (from standard or vertex form) appears in ~ 2 questions per test.
- Discriminant / number-of-solutions questions are especially common in Module 2 (hard).
- Real-world models (projectile height, revenue maximisation) appear at Medium level.
- Reflection / transformation of parabolas appears at Hard level.